

DamageScanner: A Python package for natural hazard risk assessments

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Software

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Summary

The severity and frequency of natural hazards is increasing globally (IPCC, 2023). A good understanding of the risk to buildings and assets from hazards such as extreme precipitation, record-breaking temperatures, and severe landslides is essential to develop evidence-based risk-reducing measures. To do so, we require easy-to-use and hazard-agnostic tools that allow us to compute the natural hazard risk for any place in the world. Examples are a local rainfall event, or a large-scale earthquake event. While the magnitude and intensity of such events differ greatly, the computational workflow to estimate those impacts does not. As such, the DamageScanner has been developed as a simple and ready-to-use framework to allow exposure, damage and risk assessments either through user-defined input, or simply using public data as a starting point.

Statement of need

The DamageScanner is a Python package to perform exposure, damage and risk assessments for natural hazards and climate extremes. It focusses on making the calculations easy and computationally efficient. The exposure data can be both raster- and vector-based. The DamageScanner is designed to be small, supporting both direct use and integration in larger frameworks or models, supporting researchers and hazard risk modellers. It is also easily used by students in (introductory) courses on geospatial analysis and natural hazard risk modelling. It has already been used in a number of scientific publications (Koks et al., 2019; Koks & Haer, 2020), European projects (European Commission, 2023) and integrated into other python packages and models (e.g., GEB Bruijn et al. (2023)).

State of the field

Several open-source tools exist for natural hazard risk assessments, of which the climada framework (Aznar-Siguan & Bresch, 2019; Bresch & Aznar-Siguan, 2021) is one of the most well-known open-source tools. The climada framework is a widely-used Python ecosystem to perform large-scale risk assessment for various hazards. It allows for a multitude of use cases and contains a wide range of submodules to tailor specific needs within the natural hazard risk domain (Mühlhofer et al., 2023; Riedel et al., 2024). All of which is maintained by dedicated software engineers. While the climada framework is commendable and widely used, the large ecosystem and the many dependencies within the several submodules also make it less flexible for integration within other use cases.

As such, the DamageScanner has been continuously developed independently, for several reasons. Firstly, the DamageScanner has been developed organically to tailor specific use cases within several projects. For example, the original DamageScanner was built to perform raster-based

40 damage assessments, mostly using land use information, in combination with flood hazard maps
41 (Klijn & Buren (2007); De Moel et al. (2012)). The current generation of the DamageScanner
42 has, for example, been further developed to be integrated into several infrastructure and the
43 built environment risk and resilience modelling pipelines, such as (Koks et al., 2019; Van
44 Oosterhout et al., 2023; Ye et al., 2024).

45 Secondly, with the rise of publicly-available object-based data, in particular due to
46 OpenStreetMap (OSM), a need arose to efficiently perform large-scale object-based damage
47 calculations (Koks et al., 2019; Van Ginkel et al., 2021). These object-based damage and
48 risk assessment are, however, much more computationally intensive due to the complexity of
49 exact intersections of vector and raster data. The DamageScanner fills a gap here by providing
50 an open-access approach to easily perform such risk assessment on top of existing object
51 information (see the Software Design section for an explanation of this workflow).

52 Thirdly, geospatial information of natural hazards and climate extremes are stored in a range
53 of different formats. Ranging from GeoTiffs and netCDFs, to vector-based representations of
54 earthquake intensities and flood depths. To model the impacts across those hazards within the
55 same modelling framework, a tool was needed that can readily read and use different data
56 storage formats.

57 Software design

58 The DamageScanner is implemented in Python with a core design philosophy focused on
59 simplicity. In its most simple form (as described in Figure 1), the user only needs to define
60 four key inputs: (i) vector-based objects (e.g., buildings) or a raster-based land use map, (ii)
61 a raster-based hazard-intensity map (e.g., a flood depth or wind speed), (iii) the maximum
62 damages for each of the objects or land uses, and (iv) a set of vulnerability or fragility curves
63 that describe the relation between the hazard intensity metric (e.g., the flood depth or the wind
64 speed) and the potential impact to the specific objects or land uses in the applied assessment.

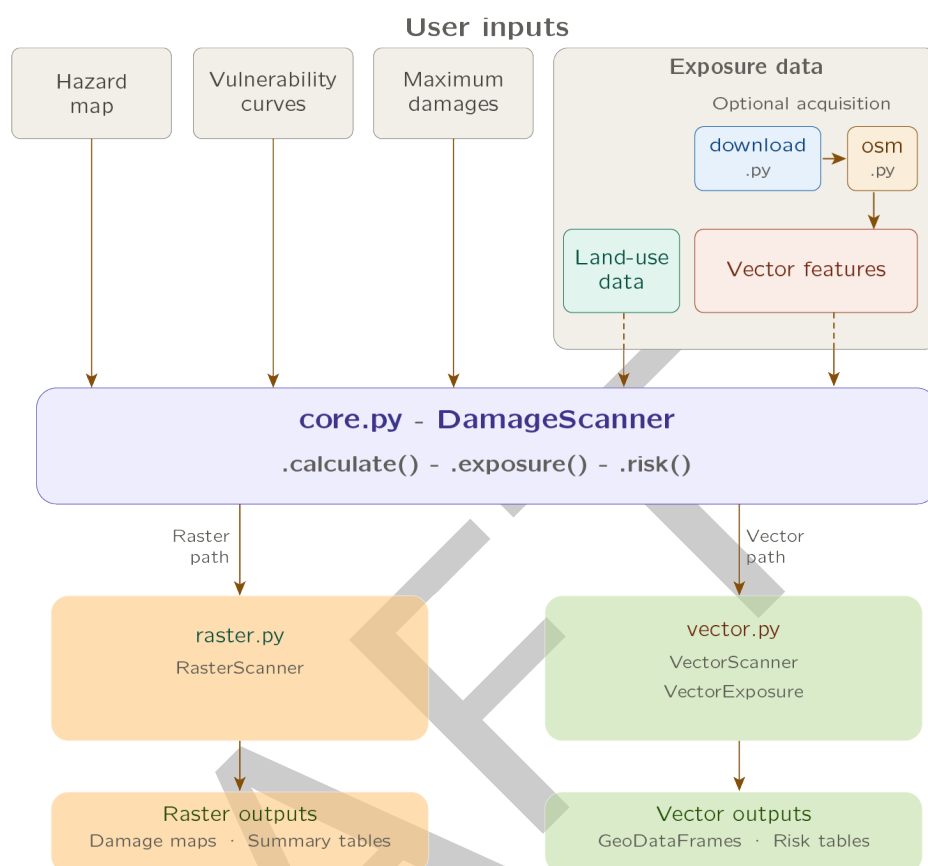


Figure 1: Software architecture of the DamageScanner package.

65 The DamageScanner can directly read most hazard maps using a provided file path (i.e., any
66 format read by rasterio or xarray like GeoTiff or NetCDF), or they can be pre-loaded through
67 the xarray package. Some examples from the public domain include JRC global flood maps
68 (Baugh et al., 2024) and USGS ShakeMap. It is recommended to clip the hazard data to the
69 area of interest before running the DamageScanner.

70 For the vulnerability curves, the DamageScanner can read all table-like data that can be read
71 by pandas, such as csv- and Excel-files. Essential is the structure of the data. The first column
72 should always represent the hazard intensity that corresponds with the hazard information
73 (e.g., centimeters of inundation, maximum gust speeds in m/s). The following columns should
74 then provide one or multiple vulnerability curves for each object/element considered in the
75 analysis (e.g., different road, building or land-use types). Most vulnerability information is
76 structured in such a manner (Nirandjan et al., 2024).

77 For the maximum damages, a file structure should be provided which links each object/element
78 within the analysis to its maximum reconstruction/replacement value. The recommended
79 structure is a left column with each object/element, and a right column with this maximum
80 damage value.

81 The final piece of input information is the exposure data, either as raster or vector. Depending
82 on the data type, the DamageScanner will perform a raster or a vector-based damage assessment.
83 Given its wide-use in OSM-based damage and risk assessments, the DamageScanner allows
84 to directly read OSM data through the download.py and osm.py scripts. Vector data can
85 contain (Multi)Polygon, (Multi)LineString or Point geometries. It is important to ensure that
86 the maximum damage values reflect these geometry types. More specifically, objects stored as
87 Polygons will require maximum damage values in m2, objects stored as LineStrings will require

88 maximum damages values in meters, whereas objects stored as Points will require maximum
89 damage values for the entire object.

90 Finally, if multiple return periods are available within the hazard data, one can estimate the
91 potential natural hazard risk, which is computed as the area under the probability-exceedence
92 loss curve through a trapezoid approach. The code-snippet below provides an example of the
93 direct use of the DamageScanner.

```
from damagescanner.core import DamageScanner

# Paths to input files
hazard = "path/to/hazard_data.tif"
feature_data = "path/to/exposure_data.shp"
curves = "path/to/vulnerability_curves.csv"
maxdam = "path/to/maxdam.csv"

# Initialize
scanner = DamageScanner(hazard, feature_data, curves, maxdam)

# Damage Assessment for single hazard
damage_results = scanner.calculate()

# Calculate risk using multiple hazard maps
hazard_dict = {
    10: "hazard_10.tif",
    50: "hazard_50.tif",
    100: "hazard_100.tif"
}

risk_results = scanner.risk(hazard_dict)

# Perform exposure-only calculation
# Curves and maxdam must be empty DataFrames for exposure-only
scanner = DamageScanner(hazard, feature_data, pd.DataFrame(), pd.DataFrame())
exposed = scanner.exposure()
```

94 When running the DamageScanner, it is important to be aware of several potential issues. The
95 first, and most common issue when running the DamageScanner is consistency in the Coordinate
96 Reference System (CRS). To reduce the risk of a potentially unsuccessful completion, it is
97 recommended to ensure that both hazard and exposure information is stored in the same CRS.
98 The second common issue relates to multiple geometries of the same object type. For example,
99 when using OSM data, school facilities can be stored as both Points or Polygons. However, the
100 maximum damages for a specific asset type can only refer to one geometry type (i.e., either
101 damages per square meter, or per entire asset). As such, one either needs to convert all assets
102 of the same object type into the same geometry type, or one should make unique object types
103 for the different geometry types (and match maximum damages accordingly).

104 Research impact statement

105 The DamageScanner has demonstrated significant research impact and grown both its user
106 base and contributor community since its initial release in 2019. It has been the basis of
107 several academic papers (e.g., Koks et al. (2019); Van Oosterhout et al. (2023); Ye et al.
108 (2024)) and European and international research projects (e.g., Horizon Europe MIRACA and
109 Ticca4Danu).

AI usage disclosure

Generative AI tools were used to support the development of the test environment for the DamageScanner (in addition to manual tests), the docstrings throughout the package (which were manually checked), and the preliminary versions of Figure 1.

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